TaskFlow Project Documentation

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1. Project Summary

TaskFlow is a monolithic full-stack task management web application built for intern learning and delivery practice. It starts with authentication and task CRUD, then expands into collaboration, analytics, teams, attachments, subtasks, time tracking, and user settings.

2. Objectives

- Provide a practical Angular + Spring Boot project for entry-level developers.
- Support task creation, assignment, tracking, and completion in a team-oriented workflow.
- Demonstrate secure authentication, clean layered architecture, and maintainable UI design.
- Add enterprise-style capabilities without microservices or external APIs.

3. Scope

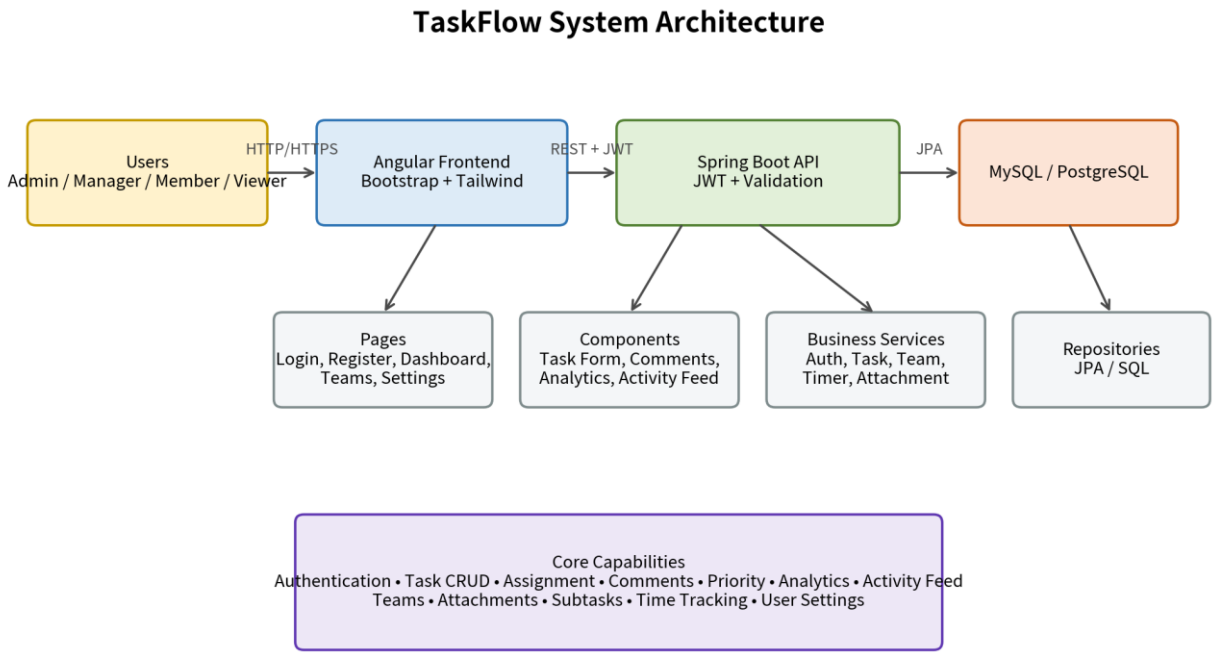
- Authentication and JWT-based access control
- Task management with status, priority, due date, and assignment
- Comments, analytics, activity feed, and due-date alerts
- Teams, file uploads, subtasks, time tracking, and settings management

4. Tech Stack

Frontend	Angular 17+, Bootstrap 5, Tailwind CSS 3
Backend	Spring Boot 3.x, Java 17+
Database	MySQL 8.x or PostgreSQL 15.x
Security	JWT, BCrypt, route guards, token blocklist
Charts	Chart.js

System Architecture

TaskFlow follows a layered monolithic design. Angular handles pages and components, Spring Boot provides REST APIs and business services, and the relational database stores operational data including tasks, comments, attachments, timers, and preferences.



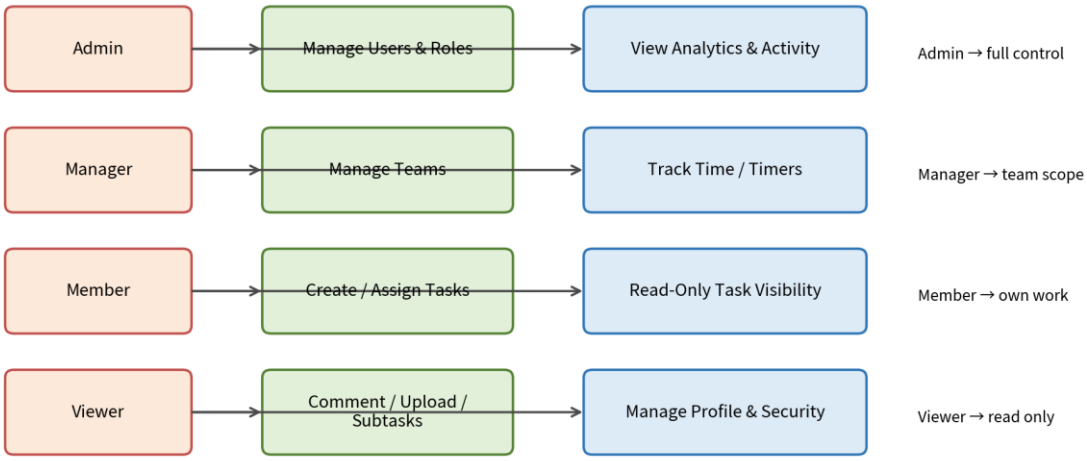
Architecture Notes

- Frontend uses page components such as Login, Register, Dashboard, Teams, and Settings.
- Backend enforces validation, JWT checks, RBAC rules, and activity logging in the service layer.
- Repositories manage persistence for users, tasks, teams, comments, attachments, timers, and preferences.

Roles and Access Model

The expanded system introduces role-based access control to support organization-wide visibility and safe delegation.

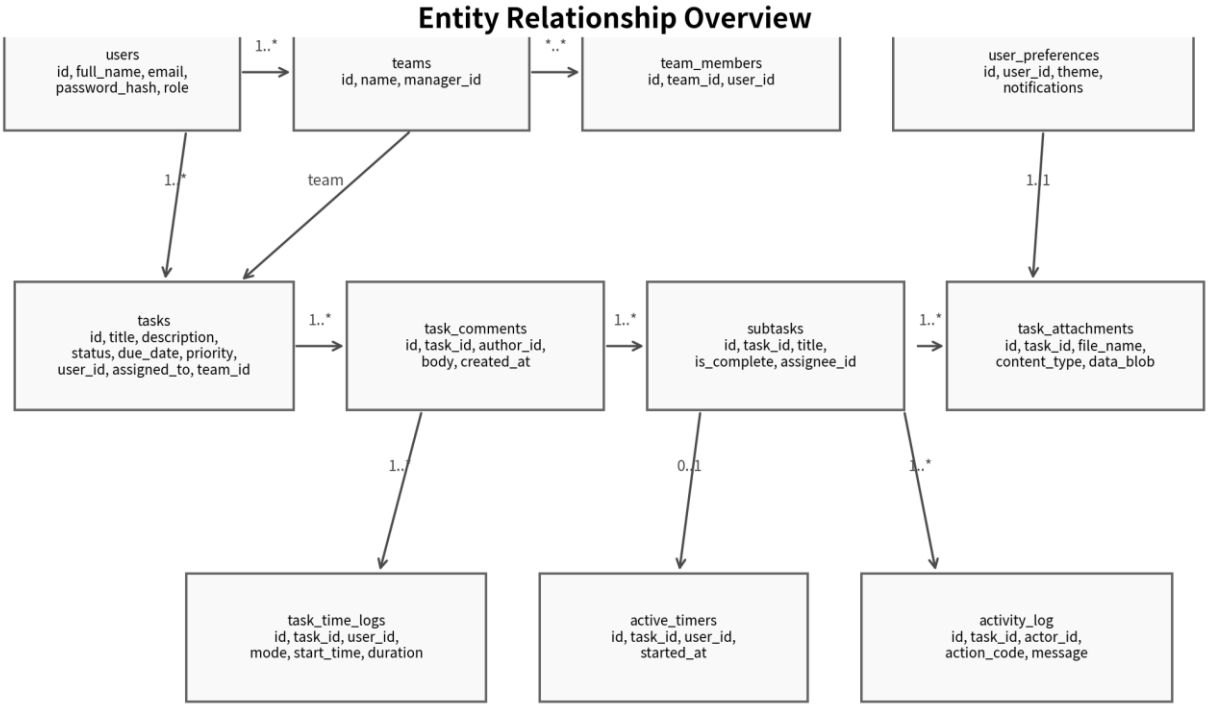
Role-Based Access and Use Cases



Capability	Admin	Manager	Member	Viewer
Manage users/roles	Yes	No	No	No
Manage teams	Yes	Yes	No	No
Create/assign tasks	Yes	Yes	Yes*	No
Comment/upload/time logs	Yes	Yes	Yes	No
View analytics/activity	Yes	Yes	Yes	Yes

Database Design

The data model extends the original users and tasks structure with collaboration, team management, attachments, timing, preferences, and security session handling.

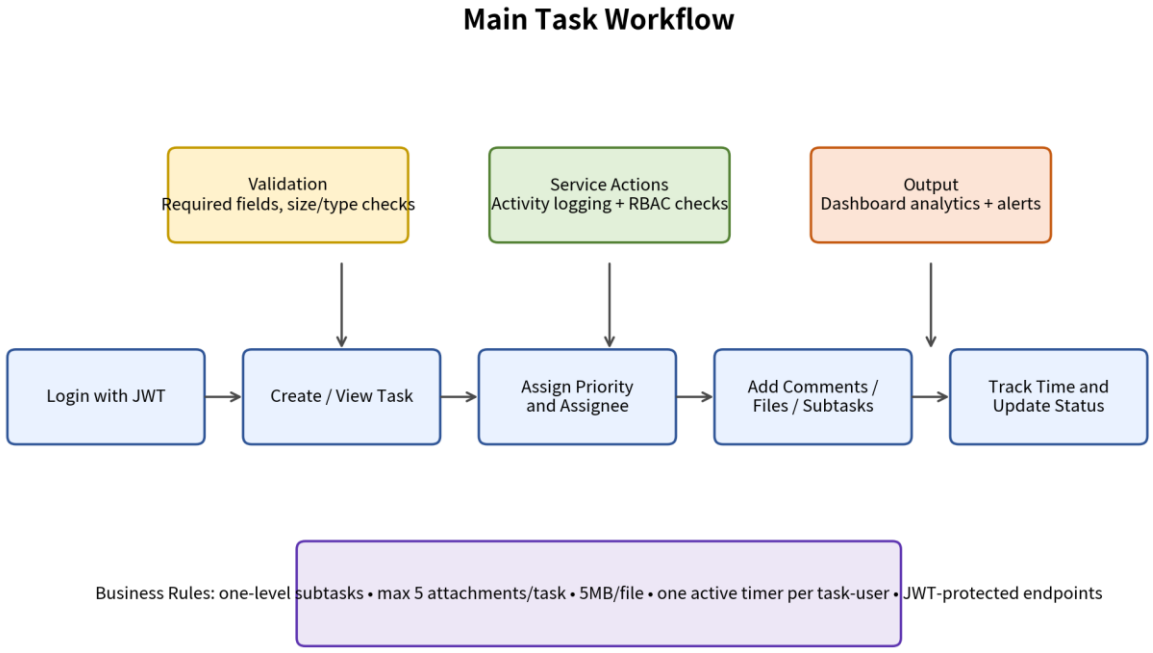


Key Entities

- users: identity, credentials, and role metadata
- tasks: core work item with owner, assignee, priority, status, due date, and optional team scope
- task_comments / activity_log: collaboration and audit trail
- task_attachments / subtasks / task_time_logs / active_timers: operational detail records

Main Workflow and Implementation Rules

The primary workflow moves from authentication into task operations, team collaboration, attachments, subtasks, and time tracking while enforcing validation and security rules.



Critical Business Rules

- Passwords are hashed with BCrypt and protected APIs require JWT authentication.
- JWT sessions can be revoked through token blocklisting.
- Comments cannot be edited; only the author may delete their own comment.
- Task priority defaults to Medium and subtasks are limited to one level.
- A task supports up to 5 attachments, each capped at 5 MB.
- Only one active timer per task-user is allowed at a time.